SOFTWARE REQUIREMENTS SPECIFICATION (IEEE STD 830-1998)

HELPBOT: AI CUSTOMER SUPPORT SYSTEM

Automated Multi-Channel Tier-1 Resolution & Telephony Deflection Platform

1. Document Control & Approvals

Project Reference	Case Study No. 97 — Regional Broadband Service Provider	**Version Baseline**	1.0 (Baseline Approved)
Governance Scope	IEEE Std 830-1998 Formatted Specification	**Target Channels**	Self-Care Web Portal & WhatsApp Messaging
Core Metrics Ceiling	Baseline Volume: 45,000 calls/month; Top 5 Drivers: 60% (27,000 calls); Target Resolution: 50% of top 5 (13,500 deflected/month)		

Role	Stakeholder Designation	Sign-off Focus	Status
Executive Sponsor	Chief Executive Officer (CEO)	Scope boundaries, SLA targets, automation mandate	Approved
Operations Sponsor	Call-Centre Operations Manager	Agent retention policies, queue deflection, handover SLA	Approved
Project Lead	Senior SEPM Lead Consultant	Traceability, engineering effort, MoSCoW prioritization	Approved
Integration Architect	Lead Systems & Billing Integrator	REST API contracts, coupling safeguards, latency limits	Approved

2. Problem Statement & Scope Management

A regional broadband provider handles **45,000 support calls per month** at a variable operational cost of **₹42 per call**. Call-centre analysis confirms that **60% of all calls (27,000 calls/month)** are concentrated entirely within five repetitive support categories: bill queries, plan changes, slow broadband speed diagnostics, regional outage checks, and payment processing.

> **Scope Governance Justification:** The CEO's expectation that HelpBot should "handle everything" is formally bounded to ensure project viability. Attempting unconstrained natural language handling across complex edge cases, hardware replacements, and legal escalations introduces uncontrollable regression. The system strictly addresses the 5 core drivers with a mandatory human handover protocol.

2.1 In-Scope vs. Out-of-Scope Boundary Matrix

Category / Query	Boundary Status	Operational Handling Rule	Impact on Time, Cost & Quality
1. Bill Queries	IN-SCOPE	Fetch invoice breakdown, unbilled usage, and PDF via Billing API.	High deflection; minimal build time via standard REST endpoints.
2. Plan Changes	IN-SCOPE	Display eligible upgrades/downgrades; execute plan switch via OTP.	Boosts ARPU; prevents misconfiguration using schema validation.
3. Slow Speed	IN-SCOPE	Execute automated ONT ping, line sync check, and router reboot advice.	Reduces longest call handling times; improves diagnostic quality.

3. Stakeholder Requirements Elicitation Synthesis

Stakeholder Group	Elicitation Technique	Key Findings & Expressed Pain Points	Requirements Derived
Broadband Customers	Questionnaires & Survey Forms	Frustrated by 15-minute IVR hold queues; demand rapid bill retrieval and self-service speed resets on WhatsApp.	FR-01, FR-02, FR-05, NFR-02
Support Agents	Contextual Inquiry & Shadowing	Agents suffer burnout from identical invoice queries; want full customer chat context preserved prior to handover.	FR-06, FR-07, NFR-07
Call-Centre Manager	Structured Semi-Structured Interview	Fears workforce downsizing and agent attrition; insists bot act as tier-1 filter, liberating agents for VIP & retention accounts.	FR-06, FR-10, Operational SLA
Chief Executive Officer	Prototype Walkthroughs	Expects broad omni-channel handling; agreed to phased deployment bounding initial release strictly to the top 5 call drivers.	FR-03, FR-04, FR-08, NFR-01

4. Functional Requirements (FR-01 to FR-10)

Req ID	Requirement Title	Atomic Functional Specification & Acceptance Criteria	MoSCoW
FR-01	Multi-Channel Ingress	The system shall ingest user messages concurrently from the self-care Web Portal and WhatsApp Business Cloud API, maintaining consistent state.	MUST
FR-02	Subscriber Authentication	The system shall verify subscriber identity using Registered Mobile Number (RMN) and a 6-digit dynamic OTP sent via SMS/WhatsApp before disclosing account data.	MUST
FR-03	Bill Inquiry & Breakdown	The system shall fetch current balance, due date, last payment details, and downloadable invoice PDFs from the Billing API via loose REST calls.	MUST
FR-04	Broadband Plan Change	The system shall retrieve eligible tariff plans, present comparisons, and record a confirmed upgrade/downgrade request with explicit user acceptance.	MUST
FR-05	Line Diagnostics (Slow Speed)	The system shall execute an automated ping test and query the Optical Line Terminal (OLT) for optical power (Rx/Tx) and profile sync speeds.	MUST
FR-06	Regional Outage Verification	The system shall check the subscriber's physical node against active network operations centre (NOC) downtime incidents and broadcast expected restoration ETAs.	MUST
FR-07	Handover Orchestration	The system shall route the chat session to a live agent queue if confidence score $C < 0.70$, if the user explicitly requests an agent, or after 2 failed retries.	MUST
FR-08	Context Transfer on Escalation	Upon handover execution, the system shall compile and inject the full conversation transcript, extracted customer intent, and account ID into the agent CRM desk.	MUST
FR-09	Payment Link Generation	The system shall generate localized, tokenized payment gateway URLs (UPI/Cards/Netbanking) and verify instant webhook status confirmation.	SHOULD

5. Non-Functional Requirements (Measurable Targets)

ID	Category	Requirement Statement & Exact Measurement Threshold	Measurement Method	MoSCoW
NFR-01	Intent Accuracy	Intent recognition precision across the 5 defined domains shall be ≥ 92.0% on standard validation benchmark sets.	Weekly confusion matrix against labeled test sets	MUST
NFR-02	Response Latency	Chatbot response generation time shall not exceed 1.8 seconds at the 95th percentile under concurrent load.	Distributed APM telemetry probes (p95)	MUST
NFR-03	System Availability	The conversational platform shall maintain ≥ 99.95% monthly uptime, excluding scheduled off-peak maintenance.	Continuous synthetic ping probes (1-minute intervals)	MUST
NFR-04	Privacy & Security	All customer PII and message payloads must be encrypted with AES-256 at rest and TLS 1.3 in transit; mask PAN/Card data to zero exposure.	Automated SAST/DAST & third-party penetration audits	MUST
NFR-05	Handover SLA	Handover routing payload shall be transferred and visible on the support agent CRM console in ≤ 3.0 seconds from trigger.	WebSocket session dispatch log timestamps	MUST
NFR-06	Billing Loose Coupling	Billing system failure or timeout (> 3.0s) shall trigger an asynchronous graceful fallback without terminating the user chat session.	Circuit breaker state monitor (Resilience4j)	SHOULD
NFR-07	Concurrency Scale	The platform shall process ≥ 1,200 concurrent active conversation sessions without degrading latency targets beyond 2.5 seconds.	LoadRunner / JMeter stress injection drills	SHOULD

6. Requirements Traceability Matrix (RTM)

Requirement ID	Requirement Title	MoSCoW	Associated Test Case ID	Design Module / UML Target
FR-01	Multi-Channel Ingress	MUST	TC-01, TC-02	ChannelWebhookController, IngressGateway
FR-02	Subscriber Authentication	MUST	TC-03, TC-04	AuthService, OTPVerificationEngine
FR-03	Bill Inquiry & Breakdown	MUST	TC-05	BillingAdapter, InvoicePDFGenerator
FR-04	Broadband Plan Change	MUST	TC-06	PlanManagementService, CatalogRepo
FR-05	Line Diagnostics (Slow Speed)	MUST	TC-07	NetworkDiagnosticsAdapter, OLTProbeClient

Requirement ID	Requirement Title	MoSCoW	Associated Test Case ID	Design Module / UML Target
FR-06	Regional Outage Verification	MUST	TC-08	NocOutageAdapter, IncidentLookupService
FR-07	Handover Orchestration	MUST	TC-09, TC-10, TC-11	HandoverRouter, AgentQueueDispatcher
FR-08	Context Transfer on Escalation	MUST	TC-12	TranscriptPackager, CrmPushService
FR-09	Payment Link Generation	SHOULD	TC-13	PaymentGatewayAdapter, WebhookListener
FR-10	Deflection & Audit Analytics	SHOULD	TC-14	AnalyticsCollector, AuditLogRepository
NFR-01 to 07	Quality Performance Targets	MUST/SHOULD	TC-15 to TC-21	Microservice Architecture & Infra Cluster